

This exam has 2 pages and 7 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

Please motivate all answers!

1. Associativity (9 points)

Is bi-implication $\leftrightarrow$ associative? If so, prove this via a truth table. If not, give a counterexample.

2. Sheffer stroke (12 points)

Express the propositional formula $\neg(\phi \oplus \psi)$ using only the Sheffer stroke $|$.

(Hint: Use the semantic equivalence $\phi \oplus \psi \equiv \neg(\phi \wedge \psi) \wedge \neg(\neg\phi \wedge \neg\psi)$.)

3. Island puzzle (12 points)

On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet islanders A , B , and C .

A says: “Either B or C is a truth speaker.” (This is an exclusive or, i.e., $\oplus$.)

B says: “ A and C are both liars.”

Determine, either through logical reasoning or via a truth table, who of A , B , and C is a truth speaker and who is a liar.

4. DPLL procedure (12 points)

Apply the DPLL procedure to the CNF $(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge \neg p$, to check whether it is satisfiable. Explain in detail how the procedure comes to a result.

5. Sets (6 + 9 points)

Consider the following set-theoretic equality

$$(A' \setminus B) \cap (C' \setminus B) = (A \cup C)' \setminus B.$$

(a) Using the method you have learned in the lectures, for each side of the equality, draw a sequence of Venn diagrams that clearly illustrates how the set is constructed from A , B and C using the fundamental set operations, and use this to show that the set-theoretic equality holds for all sets A , B and C .

(b) Prove the set-theoretic equality using the rules of the algebra of sets.

6. Relations (6 + 8 points)

Suppose R and S are relations of type $\{1, 2, 3, 4\} \times \{a, b, c, d\}$ defined by

$$R := \{\langle 1, a \rangle, \langle 2, a \rangle, \langle 2, b \rangle, \langle 3, b \rangle, \langle 3, c \rangle, \langle 4, c \rangle\},$$

$$S := \{\langle 1, b \rangle, \langle 2, d \rangle, \langle 3, a \rangle, \langle 4, b \rangle, \langle 4, d \rangle\}.$$

- Write down the matrix representations of the relations R and S .
- Draw a Venn diagram depicting the composition $R^{-1} \circ S$, and explicitly list all the elements of the composite relation $R^{-1} \circ S$ in the curly-bracket notation.

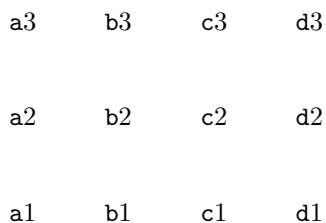
7. Ordering relation (8 + 8 points)

Suppose that in each of the sets $A := \{a, b, c, d\}$ and $B := \{1, 2, 3\}$ we have a partial ordering relation that is described by, respectively, the left or right Hasse diagram shown below:



These partial ordering relations in A and B induce a lexicographic order and a cartesian order on the product set $A \times B$.

- Draw the Hasse diagram for the lexicographic order on $A \times B$ by adding arrows to the diagram you see below (where $a1$ is short for $\langle a, 1 \rangle$, et cetera).



NB: you do *not* have to use the algorithm you learned to construct the Hasse diagram and you do *not* have to explain how you obtained your diagram.

- Likewise, draw the Hasse diagram for the cartesian order on $A \times B$.